Probabilistic Combinatorics (Math 5390) Fall 2026
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Exercise Sheet 2

Due date: 12:20, Oct 23rd.
Late submissions will receive no credit.

You and your partner should try to solve all of the exercises below, and then submit a total of two solutions to be graded — each problem is worth 10 points. Both partners should author one solution each, making clear who wrote which solution, but you will earn credit for both solutions. Make sure to clearly justify your answers, defining everything precisely and stating any results you are using. You should then submit your solutions on NTU COOL, either as a typed PDF or a clear picture or scan of your handwritten answers.

Exercise 1 Given natural numbers $a_1, a_2, \dots, a_r$ and $b_1, b_2, \dots, b_s$, we denote by $\mathcal{L}$ the linear equation $\sum_{i=1}^{r} a_i x_i = \sum_{j=1}^{s} b_j y_j$, and say a set of integers A is *$\mathcal{L}$-free* if it does not contain any solution $(x_1, \dots, x_r, y_1, \dots, y_s) \in A^{r+s}$ to $\mathcal{L}$. Prove that if $\sum_{i=1}^{r} a_i \neq \sum_{j=1}^{s} b_j$, then there is some constant $c = c_{\mathcal{L}}$ such that every set S of n non-zero integers contains an $\mathcal{L}$-free subset $A \subseteq S$ of size $|A| \geq cn$.

Solution. Write $\sigma_a = \sum_{i=1}^{r} a_i$ and $\sigma_b = \sum_{j=1}^{s} b_j$. A set contains a solution to $\mathcal{L}$ if and only if it contains a solution to the equation obtained by swapping the two sides of $\mathcal{L}$, so we may assume $\sigma_a > \sigma_b$; as these are integers, $\sigma_a - \sigma_b \geq 1$. For $u \in \mathbb{R}$ let $\{u\} = u - \lfloor u \rfloor \in [0, 1)$ denote its fractional part. Set

$$t = \frac{1}{2(\sigma_a - \sigma_b)}, \qquad \delta = \frac{1}{4(\sigma_a + \sigma_b)}, \qquad I = [t - \delta, t + \delta), \qquad c = 2\delta = \frac{1}{2(\sigma_a + \sigma_b)}.$$

Since $0 < \sigma_a - \sigma_b \leq \sigma_a + \sigma_b$, we have $t - \delta \geq \frac{1}{2(\sigma_a+\sigma_b)} - \frac{1}{4(\sigma_a+\sigma_b)} > 0$ and $t + \delta \leq \frac{1}{2} + \frac{1}{4} < 1$, so $I \subseteq (0, 1)$ is an interval of length c. For $\theta \in [0, 1)$ let

$$A_\theta = \{x \in S : \{x\theta\} \in I\}.$$

Claim 1: A_θ is $\mathcal{L}$-free for every θ. Suppose $(x_1, \dots, x_r, y_1, \dots, y_s) \in A_\theta^{r+s}$ solves $\mathcal{L}$, and put $u = \sum_i a_i \{x_i\theta\}$ and $v = \sum_j b_j \{y_j\theta\}$.

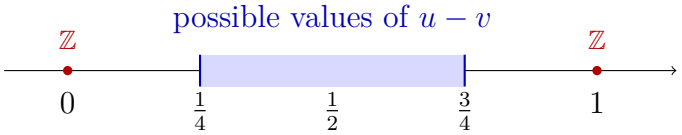
$u - v$ is an integer. Each $x_i\theta - \{x_i\theta\}$ and $y_j\theta - \{y_j\theta\}$ is an integer, and the a_i, b_j are integers, so

$$u - v \equiv \theta\Big(\sum_i a_i x_i - \sum_j b_j y_j\Big) = 0 \pmod 1.$$

$u - v \in [\frac{1}{4}, \frac{3}{4}]$. Each $\{x_i\theta\}, \{y_j\theta\}$ lies in $[t - \delta, t + \delta]$ and all $a_i, b_j \geq 0$, so $\sigma_a(t - \delta) \leq u \leq \sigma_a(t + \delta)$ and $\sigma_b(t - \delta) \leq v \leq \sigma_b(t + \delta)$. Hence

$$u - v \geq (\sigma_a - \sigma_b)t - (\sigma_a + \sigma_b)\delta = \tfrac{1}{2} - \tfrac{1}{4} = \tfrac{1}{4}, \qquad u - v \leq (\sigma_a - \sigma_b)t + (\sigma_a + \sigma_b)\delta = \tfrac{3}{4}.$$

As $[\frac{1}{4}, \frac{3}{4}]$ contains no integer, this is a contradiction. □



Claim 2: if θ is uniform on $[0,1)$ and $x \neq 0$ is an integer, then $\Pr(\{x\theta\} \in I) = c$. Let $m = |x| \geq 1$ and split $[0,1)$ into the m intervals $J_k = [\frac{k}{m}, \frac{k+1}{m})$, $0 \leq k \leq m-1$. On each J_k, $x\theta$ runs over an interval of length 1 between two consecutive integers, so $\theta \mapsto \{x\theta\}$ is an affine map with slope $\pm m$ from J_k onto $[0,1)$ (up to an endpoint). Hence $\{\theta \in J_k : \{x\theta\} \in I\}$ has length $|I|/m = c/m$, and summing over the m intervals gives $\Pr(\{x\theta\} \in I) = c$. □

Conclusion. Choose θ uniformly at random from $[0,1)$. Every element of S is a non-zero integer, so by linearity of expectation and Claim 2,

$$\mathbb{E}\,|A_\theta| = \sum_{x \in S} \Pr(\{x\theta\} \in I) = cn.$$

Hence some θ satisfies $|A_\theta| \geq cn$, and by Claim 1, $A = A_\theta \subseteq S$ is $\mathcal{L}$-free. Thus $c_{\mathcal{L}} = \frac{1}{2(\sigma_a + \sigma_b)}$ works. □

Exercise 2 Recall that a k-uniform family of subsets $\mathcal{F} \subseteq \binom{[n]}{k}$ is said to be *intersecting* if $F_1 \cap F_2 \neq \emptyset$ for all $F_1, F_2 \in \mathcal{F}$. We say that $\mathcal{F}$ is *maximally intersecting* if it is intersecting and, if $\mathcal{F} \subsetneq \mathcal{F}'$, then $\mathcal{F}'$ is not intersecting. Given a family $\mathcal{G} \subseteq \binom{[n]}{k}$ of sets, we write $\mathcal{I}(\mathcal{G}) = \left\{F \in \binom{[n]}{k} : \forall G \in \mathcal{G},\ F \cap G \neq \emptyset\right\}$ for the family of all k-sets that intersect every member of $\mathcal{G}$. We say a maximally intersecting family $\mathcal{F}$ is *generated* by $\mathcal{G}$ if $\mathcal{I}(\mathcal{G}) = \mathcal{F}$.

(a) Show that $\mathcal{F}$ is intersecting if and only if $\mathcal{F} \subseteq \mathcal{I}(\mathcal{F})$, and it is maximally intersecting if and only if we have equality.

(b) Prove that every maximally intersecting family $\mathcal{F}$ is generated by a family $\mathcal{G}$ with $|\mathcal{G}| \leq \binom{2k}{k}$.

Solution. Throughout $k \geq 1$, so every $F \in \binom{[n]}{k}$ satisfies $F \cap F = F \neq \emptyset$. Note that $\mathcal{I}$ is order-reversing: if $\mathcal{G}_1 \subseteq \mathcal{G}_2$, then $\mathcal{I}(\mathcal{G}_2) \subseteq \mathcal{I}(\mathcal{G}_1)$, since a k-set meeting every member of $\mathcal{G}_2$ in particular meets every member of $\mathcal{G}_1$.

Proof of (a). By definition, $\mathcal{F} \subseteq \mathcal{I}(\mathcal{F})$ means that every $F_1 \in \mathcal{F}$ meets every $F_2 \in \mathcal{F}$, which is exactly the statement that $\mathcal{F}$ is intersecting.

Now suppose $\mathcal{F}$ is maximally intersecting; then $\mathcal{F} \subseteq \mathcal{I}(\mathcal{F})$. If there were some $H \in \mathcal{I}(\mathcal{F}) \setminus \mathcal{F}$, then $\mathcal{F}' = \mathcal{F} \cup \{H\} \supsetneq \mathcal{F}$ would be intersecting: H meets every member of $\mathcal{F}$ since $H \in \mathcal{I}(\mathcal{F})$, $H \cap H \neq \emptyset$, and any two members of $\mathcal{F}$ meet. This contradicts maximality, so $\mathcal{F} = \mathcal{I}(\mathcal{F})$.

Conversely, suppose $\mathcal{F} = \mathcal{I}(\mathcal{F})$. Then $\mathcal{F}$ is intersecting by the first part. Let $\mathcal{F}' \supsetneq \mathcal{F}$ and pick $H \in \mathcal{F}' \setminus \mathcal{F}$. If $\mathcal{F}'$ were intersecting, H would meet every member of $\mathcal{F} \subseteq \mathcal{F}'$, i.e. $H \in \mathcal{I}(\mathcal{F}) = \mathcal{F}$, a contradiction. So $\mathcal{F}$ is maximally intersecting. □

We use the following result from lecture.

Lemma (Bollobás's set-pairs inequality)**.** *Let* $A_1, \dots, A_m, B_1, \dots, B_m \in \binom{[n]}{k}$ *satisfy* $A_i \cap B_i = \emptyset$ *for all* i*, and* $A_i \cap B_j \neq \emptyset$ *for all* $i \neq j$*. Then* $m \leq \binom{2k}{k}$.

Proof of (b). Let $\mathcal{F}$ be maximally intersecting. By (a), $\mathcal{I}(\mathcal{F}) = \mathcal{F}$, so $\mathcal{F}$ is generated by the subfamily $\mathcal{F}$ of itself. Among all subfamilies $\mathcal{G} \subseteq \mathcal{F}$ with $\mathcal{I}(\mathcal{G}) = \mathcal{F}$, choose one of minimum size, say $\mathcal{G} = \{G_1, \dots, G_m\}$. We show $m \leq \binom{2k}{k}$; if $m = 0$ there is nothing to prove.

Fix i. By minimality, $\mathcal{I}(\mathcal{G} \setminus \{G_i\}) \neq \mathcal{F} = \mathcal{I}(\mathcal{G})$, while $\mathcal{I}(\mathcal{G}) \subseteq \mathcal{I}(\mathcal{G} \setminus \{G_i\})$ as $\mathcal{I}$ is order-reversing. So there exists

$$H_i \in \mathcal{I}(\mathcal{G} \setminus \{G_i\}) \setminus \mathcal{I}(\mathcal{G}).$$

Since $H_i \in \mathcal{I}(\mathcal{G} \setminus \{G_i\})$, it meets G_j for every $j \neq i$. Since $H_i \notin \mathcal{I}(\mathcal{G})$, it is disjoint from some member of $\mathcal{G}$, which can only be G_i. Thus

$$G_i \cap H_i = \emptyset \quad \text{for all } i, \qquad G_j \cap H_i \neq \emptyset \quad \text{for all } i \neq j.$$

Applying the Lemma with $A_i = G_i$ and $B_i = H_i$ gives $|\mathcal{G}| = m \leq \binom{2k}{k}$, and $\mathcal{G}$ generates $\mathcal{F}$. □

Exercise 3 Recall the asymmetric Ramsey number $R(k, \ell)$ is the smallest n such that every red-/blue-edge-colouring of K_n contains a red K_k or a blue K_ℓ.

(a) Show that for any integer n and $p \in [0, 1]$, we have

$$R(k, \ell) > n - \binom{n}{k} p^{\binom{k}{2}} - \binom{n}{\ell} (1-p)^{\binom{\ell}{2}}.$$

(b) Use the result from (a) to prove the best lower bound on $R(3, \ell)$ (as $\ell \to \infty$) that you can.

Solution. Throughout, ln is the natural logarithm and $k, \ell \geq 1$.

Proof of (a). If $n \leq 0$, the right-hand side is at most $n \leq 0 < R(k, \ell)$, so assume $n \geq 1$. Colour each edge of K_n red with probability p and blue otherwise, independently. Let X be the number of red copies of K_k and Y the number of blue copies of K_ℓ. A fixed k-set of vertices spans a red K_k exactly when all of its $\binom{k}{2}$ edges are red, so by linearity of expectation

$$\mathbb{E}[X + Y] = \binom{n}{k} p^{\binom{k}{2}} + \binom{n}{\ell} (1-p)^{\binom{\ell}{2}}.$$

Hence there is a colouring with $X+Y \leq \mathbb{E}[X+Y]$. Fix such a colouring, and delete one vertex from each red K_k and each blue K_ℓ. This leaves a set W of $m \geq n - X - Y \geq n - \mathbb{E}[X+Y]$ vertices, and the colouring on W has no red K_k and no blue K_ℓ, since each such copy lost

a vertex. Restricting to any $m' \le m$ vertices of W gives the same property, so no $n' \le m$ satisfies the defining property of $R(k,\ell)$. Thus

$$R(k,\ell) > m \ge n - \binom{n}{k} p^{\binom{k}{2}} - \binom{n}{\ell}(1-p)^{\binom{\ell}{2}}. \qquad \square$$

Proof of (b). We show $R(3,\ell) = \Omega\big((\ell/\ln\ell)^{3/2}\big)$. Fix $p \in (0,1]$. Using $\binom{n}{3} \le \frac{n^3}{6}$, $\binom{n}{\ell} \le \big(\frac{en}{\ell}\big)^\ell$ and $1-p \le e^{-p}$, (a) gives for every n

$$R(3,\ell) > \underbrace{n - \frac{p^3 n^3}{6}}_{=:h(n)} - \Big(\underbrace{\frac{en}{\ell} e^{-p(\ell-1)/2}}_{=:b(n,p)}\Big)^\ell. \qquad (\star)$$

Step 1: choosing n in terms of p. Since $h'(x) = 1 - \frac{p^3x^2}{2}$, h increases on $[0,x^*]$ and decreases afterwards, where $x^* = \sqrt{2}\,p^{-3/2}$ solves $h'(x^*) = 0$; so the best choice is $n \approx x^*$. Take $n = \lfloor x^* \rfloor$, so $x^* - 1 < n \le x^*$. From $n \le x^* = \sqrt{2}\,p^{-3/2}$ we get $p^3 n^2 \le 2$, hence

$$h(n) = n\left(1 - \frac{p^3n^2}{6}\right) \ge n\left(1 - \frac{2}{6}\right) = \frac{2n}{3} > \frac{2}{3}(x^* - 1) = \frac{2\sqrt{2}}{3}p^{-3/2} - \frac{2}{3}.$$

Step 2: choosing p. By Step 1, the main term of $(\star)$ is $F(p) = \frac{2\sqrt{2}}{3}p^{-3/2}$, and $F'(p) = -\sqrt{2}\,p^{-5/2} < 0$, so we want p as small as possible. The only constraint is that the blue term should be negligible, i.e. $b(n,p)^\ell = o(1)$ as $\ell \to \infty$. Try $p = \frac{c\ln\ell}{\ell-1}$ with $c > 0$. Then $e^{-p(\ell-1)/2} = \ell^{-c/2}$ and $p^{-1} = \frac{\ell-1}{c\ln\ell} \le \frac{\ell}{c\ln\ell}$ (as $\ell - 1 \le \ell$), so using $n \le \sqrt{2}\,p^{-3/2}$,

$$b(n,p) \le \frac{e\sqrt{2}}{\ell}\left(\frac{\ell}{c\ln\ell}\right)^{3/2} \ell^{-c/2} = \frac{e\sqrt{2}}{c^{3/2}} \cdot \frac{\ell^{(1-c)/2}}{(\ln\ell)^{3/2}}.$$

This tends to 0 exactly when $c \ge 1$, and then $b(n,p)^\ell \le b(n,p) = o(1)$. As F is decreasing, we take the smallest such value $c = 1$, i.e.

$$p = \frac{\ln\ell}{\ell - 1} \in (0,1], \qquad b(n,p) \le \frac{e\sqrt{2}}{(\ln\ell)^{3/2}}.$$

Conclusion. For $\ell \ge 12$ we have $(\ln\ell)^{3/2} > e\sqrt{2}$, so $b(n,p) < 1$ and $b(n,p)^\ell \le b(n,p) = o(1)$. By $(\star)$ and Step 1,

$$R(3,\ell) > \frac{2\sqrt{2}}{3}p^{-3/2} - \frac{2}{3} - o(1) = \frac{2\sqrt{2}}{3}\left(\frac{\ell-1}{\ln\ell}\right)^{3/2} - O(1) = \Omega\left(\left(\frac{\ell}{\ln\ell}\right)^{3/2}\right).$$

Step 1 shows that (a) never gives more than $O(p^{-3/2})$, and Step 2 that the blue term forces $p = \Omega\big(\frac{\ln\ell}{\ell}\big)$, so this is the best bound obtainable from (a), up to the constant. $\square$

Exercise 4 In this exercise you will provide an explicit construction of a set of n points in $[0,1]^2$, such that every triple of points determines a triangle of area $\Omega\left(\frac{1}{n^2}\right)$.

(a) Let p be prime, and let $S = \{(x, x^2 \pmod p) : 0 \le x \le p-1\}$. Prove that there are no three points of S lying on a common line.

(b) By rescaling S appropriately, show that we can find p points in $[0,1]^2$, such that every triangle has area at least $\frac{1}{2(p-1)^2}$.

(c) Deduce the existence of the desired set of points for all $n \ge 2$.

Solution. Throughout, $x^2 \pmod p$ denotes the representative in $\{0, 1, \dots, p-1\}$, so $S \subseteq \{0, 1, \dots, p-1\}^2 \subseteq \mathbb{Z}^2$. For points $P_i = (x_i, y_i) \in \mathbb{R}^2$, $i = 1, 2, 3$, write

$$D(P_1, P_2, P_3) = \det \begin{pmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{pmatrix}.$$

Recall that the triangle $P_1P_2P_3$ has area $\frac{1}{2}|D(P_1, P_2, P_3)|$, and in particular P_1, P_2, P_3 are collinear if and only if $D(P_1, P_2, P_3) = 0$.

Proof of (a). Take three distinct points of S. Each has the form $P_i = (x_i, y_i)$ with $x_i \in \{0, 1, \dots, p-1\}$ and $y_i = x_i^2 \pmod p$, for $i = 1, 2, 3$. If $x_i = x_j$, then also $y_i = y_j$, so $P_i = P_j$; hence x_1, x_2, x_3 are pairwise distinct. It suffices to show that $D = D(P_1, P_2, P_3) \neq 0$.

Since y_i is the remainder of x_i^2 upon division by p, there are integers q_i with $y_i = x_i^2 - pq_i$ for $i = 1, 2, 3$. Thus the third column of the matrix defining D is $(x_i^2)_i - p\,(q_i)_i$, and since the determinant is linear in its third column,

$$D = \underbrace{\det \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix}}_{=:V} - p \underbrace{\det \begin{pmatrix} 1 & x_1 & q_1 \\ 1 & x_2 & q_2 \\ 1 & x_3 & q_3 \end{pmatrix}}_{=:E}.$$

Both V and E are determinants of integer matrices, hence integers, so $D \equiv V \pmod p$. Moreover V is a Vandermonde determinant, so

$$V = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2).$$

As $0 \le x_i \le p-1$ are distinct, each difference $x_j - x_i$ satisfies $0 < |x_j - x_i| < p$, so is not divisible by p. Since p is prime, the product V is not divisible by p either. Hence $D \equiv V \not\equiv 0 \pmod p$; in particular $D \neq 0$, and P_1, P_2, P_3 are not collinear. □

Proof of (b). For $p = 2$ there are no triangles, so assume $p \ge 3$. Let $S' = \frac{1}{p-1}S = \left\{\left(\frac{x}{p-1}, \frac{x^2 \pmod p}{p-1}\right) : 0 \le x \le p-1\right\}$. Since all coordinates of points of S lie in $\{0, \dots, p-1\}$, we have $S' \subseteq [0,1]^2$, and $|S'| = |S| = p$.

Take three distinct points of S'; they are $\frac{1}{p-1}P_1, \frac{1}{p-1}P_2, \frac{1}{p-1}P_3$ for distinct $P_1, P_2, P_3 \in S$. Let $D = D(P_1, P_2, P_3)$. We claim $|D| \ge 1$:

- $D \in \mathbb{Z}$: every entry of the matrix defining D is an integer (the coordinates of points of S are integers), and a determinant is a sum of products of entries with signs ± 1, so it is an integer.
- $D \neq 0$: this was shown in the proof of (a).

A non-zero integer has absolute value at least 1, so $|D| \geq 1$, and the triangle $P_1P_2P_3$ has area $\frac{1}{2}|D| \geq \frac{1}{2}$.

Finally, multiplying the second and third columns of the matrix by $\frac{1}{p-1}$ multiplies the determinant by $\frac{1}{(p-1)^2}$, so

$$D\left(\tfrac{1}{p-1}P_1, \tfrac{1}{p-1}P_2, \tfrac{1}{p-1}P_3\right) = \frac{D}{(p-1)^2},$$

and the triangle with vertices $\frac{1}{p-1}P_i$ has area $\frac{|D|}{2(p-1)^2} \geq \frac{1}{2(p-1)^2}$. □

Proof of (c). If $n = 2$, any two distinct points of $[0,1]^2$ work: there is no triple of points, hence no triangle, so there is nothing to check. Now let $n \geq 3$. By Bertrand's postulate (for every integer $m > 1$ there is a prime p with $m < p < 2m$), there is a prime p with $n < p < 2n$. Let T be any n points of the set S' from (b). Then $T \subseteq [0,1]^2$, and every triangle with vertices in T is a triangle with vertices in S', so its area is at least

$$\frac{1}{2(p-1)^2} > \frac{1}{2(2n)^2} = \frac{1}{8n^2}.$$

Hence every triple of points of T determines a triangle of area $\Omega\left(\frac{1}{n^2}\right)$. □